

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<thinking process >

Step 1 : Calculate the distance John travels during the initial 3 hours .

He drives at 60 mph for 3 hours .

So , distance = speed × time = 60 mph × 3 hours = 180 miles .

So , he is 180 miles from home after the first 3 hours .

Step 2 : Understand that he turns around and tries to get home in 4 hours .

He spends the first 2 hours in standstill traffic → so he does not move during those 2 hours .

Then , he drives for half an hour (0 . 5 hours) at 30 mph .

Distance covered in this leg = 30 mph × 0 . 5 hours = 15 miles .

Then , he drives the rest of the time (4 hours total - 2 hours - 0 . 5 hours) = 1 . 5 hours at 80 mph .

Distance covered = 80 mph × 1 . 5 hours = 120 miles .

Total distance traveled toward home during the 4 hours = 15 + 120 = 135 miles .

Step 3 : Since he started 180 miles away from home and moved 135 miles toward home , his distance from home at the end is :

Final distance = Initial distance - Distance covered toward home = 180 - 135 = 45 miles .

So , he is 45 miles from home at the end of those 4 hours .

< answer > 45 miles </answer >

